DIAL-Det: Appendices

Technical appendices to the IEEE ICASSP paper
"DIAL-Det: learning identity mass for budgeted detection allocation"

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https://github.com/VKHS/Dial-Det

Scope. These appendices complete the four-page paper [1]. Appendix A positions DIAL-Det against every family of detection decoders. Appendix B specifies the evidence model to the level required for reimplementation. Appendix C proves every analytic statement of the paper and adds four results: the matroid exchange behind the cost of a retained duplicate (Lemma C1), the complete admission cut-off (Proposition C1), a closed-form exchange boundary (Proposition C2) and the soundness of the executed arithmetic (Proposition C3); it proves the matching audit (Theorem C1) and the risk guarantee of the deployed policy (Theorem C2). Appendix D specifies the solver, its certificate record and its failure policy; Appendix E defines every reported quantity with its population and weighting, together with the complete measurement protocol; Appendix F reports extended results; Appendix G reproduces the pre-registered analysis specification; Appendix H documents the provenance of every number.

Numbering. Unlettered numbers — equations (1)–(9), Propositions 1–2, Theorem 1, Algorithm 1, Figures 1–5, Tables 1–2 and Sections 1–3 — refer to the paper; lettered numbers refer to these appendices, and the map below locates each statement of the paper.

Values. Every experimental number is generated from the recorded evaluation outputs by `tools/dial_values.py`, which writes the data blocks of the paper and of these appendices; numbers shared with the paper are identical by construction. The appendix-specific numerical entries and evaluated settings below are the recorded values used by the reported analyses. Analytic panels, worked examples and closed forms are exact.

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Notation

Symbol	Meaning	Paper
V, N; G	frozen candidates of an image, $N = \lvert V\rvert \le 500$; its non-ignored ground-truth objects	Sec. 1
K; S	report budget; report $S \subseteq V$ with $\lvert S\rvert \le K$	Sec. 1
$\nu_\tau(A,G)$	size of a maximum class-consistent matching between A and G at IoU $\ge \tau$	Sec. 1
H, Q, C, C_{ev}	$\nu_\tau(V,G)$; $\min(K,H)$; $\nu_\tau(S,G)$; matches of the evaluator's score-ordered assignment	(1)
q_i; r_{ij}	quality: probability that i matches its object at the evaluator's IoU; identity affinity, $r_{ii} = 1$	Sec. 2
$s, \widetilde{r}$; θ, p	$s = r^p$; retained affinity $\widetilde{r} = s\,\mathbf{1}[s \ge \theta]$; retention threshold and sharpening exponent	(3)
E, $E(u)$; E_o, E_r	support graph and the support neighbours of u; its overlap and rescue branches	Sec. 2
d_u, $d_u^\star$, e_u	identity mass $\sum_j r_{ju}^p$; target mass; log-mass error $\log d_u - \log d_u^\star$	(2), Prop. 1
μ_u; D_B	demand weight $1/d_u$; block demand $\sum_{u\in B}\mu_u$	(3)
$m_u(S)$; w_{ij}	coverage $\max_{i\in S} q_i \widetilde{r}_{iu}$; pair weight $\widetilde{r}_{ij}\min(q_i,q_j)$	(3)
ρ, η, λ	price per box, coverage weight, redundancy weight	(3)
F; $D_j(S)$; $X_{i\to j}$	objective; marginal gain of $j \notin S$; exchange gain	(3), (6)
δ; F_δ; B_δ, L_δ; g_F, ε_g	grid 2^{-30}; objective on grid coefficients; certified upper bound, incumbent value; certified gap, grid error	(8)
R_w, R_a; U; J, h, α	per-image waste and shortfall; empirical-Bernstein bound; family size, $\log(2J/\alpha)$, level	(9)
L, G, S (decoders)	tuned local rule; greedy maximization of F; Algorithm 1	Secs. 2–3

Map from the paper

Statement of the paper	Where it is proved or extended
Decomposition (1); on a full report every retained duplicate costs one attainable instance (Sec. 1)	Lemma C1
F is submodular, can be negative and has degree three, so no QUBO represents it (Sec. 2)	Lemma C2
Proposition 1: identity mass, expected matches (4), mass-error bound (5)	Definition C1, App. C.2
Fragmentation and bridging, the two failures beyond the bound (Sec. 2)	Examples C1–C2
Proposition 2(a): admission cut-off (7) and its two further branches	Proposition C1, Fig. C1a,b
Proposition 2(b): exchange boundary that no keep-below threshold reproduces	Proposition C2, Corollary C1, Fig. C1c,d
Theorem 1: checked gap (8) under directed rounding	Proposition C3, App. C.7
Algorithm 1: checked allocation	Algorithm D1, Tables D1 and F2
Risk bound (9) and the selection of (ρ,λ)	Theorem C2, Table F1, Fig. F2a
Miss bound quoted in Sec. 3	Theorem C1, Table F2
Table 1: sixteen settings, Holm families, seeds	Fig. F1, Tables E3 and F6
Figs. 3–4: factorial, removals, oracle, heterogeneity, identity evidence	App. E, Table F4, Fig. F3a
Fig. 5 and the budget paragraph	App. E.7, Table F5
Decoding time	Table F3, Fig. F3b
Pre-registered plan	App. G

Appendix A Positioning against decoder families

Table A1 places DIAL-Det among the decoder families of the paper's positioning paragraph (Sec. 1). Entries describe each family's published decision rule at the level of the family; individual methods may add a component, such as a learned kernel or an extra head. No entry is a measured result.

Suppression rules. Greedy NMS and its tuned variants [3, 4] keep a box when its overlap with every kept box is below a threshold; learned variants [5, 6, 7, 8] replace the overlap test by a network score, spatially adaptive suppression [9] adjusts the threshold from spatial context without training, and crowd variants [10, 11, 12, 13, 14] adapt it through density,

Table A1. Decoder families against the properties DIAL-Det combines. *Unit*: what the objective or rule values; *cap*: whether the report budget enters the decision ("after": top-K truncation of the output); *pair evidence*: what decides whether two boxes show one object; *transitivity*: how same-identity consistency is enforced; *guarantee*: what the solver certifies; *risk control*: what is certified about the output; *detector*: whether the detector is left unchanged.

Family	unit	cap	pair evidence	transitivity	guarantee	risk control	detector
Rule and spatially adaptive suppression [3, 4, 9]	box	after	IoU, DIoU, spatial context	–	rule	–	frozen
Learned suppression [5, 6, 7, 8]	box	after	learned	–	rule	–	frozen or joint
IoU-aware calibration [31]	box	after	learned, post hoc	–	rule	–	frozen
Crowd suppression [10, 11, 12, 13, 14]	box	after	density, visibility	–	rule	–	extra heads
Quadratic and QUBO selection [15, 16, 17]	candidate	after	overlap, appearance	–	heuristic	–	frozen
Determinantal selection [18, 19, 20]	candidate	after	kernel	–	greedy MAP	–	frozen
Facility location [21]	hypothesis	per-box cost	Hough votes	votes partitioned	greedy	–	frozen
Correlation clustering, multicut [22, 23]	identity	–	learned	hard, at inference	LP bound or heuristic	–	frozen
One-to-one and distinct queries [24, 25, 26, 27, 28, 29]	query	query count	implicit	–	–	–	retrained
One-to-many training with NMS [30]	box	after	IoU	–	rule	–	retrained
Conformal detection [32, 33, 34]	box	–	–	–	–	box coverage	frozen
DIAL-Det	identity mass	$\lvert S\rvert \le K$, price ρ	two learned branches	trained ($\mathscr{L}_t$)	checked gap (Thm. 1)	waste, shortfall	frozen

visibility or representative-region cues, mostly from heads trained with the detector. All of them decide pairs in score order and apply the cap afterwards. None can trade a retained duplicate for a distinct object once the report is full, and Corollary C1 proves that no keep-below rule reproduces that exchange. IoU-aware calibration [31] models duplication in post-hoc confidences and leaves selection to the ranking, so it cannot make the trade either.

Global selectors. Quadratic and QUBO formulations [15, 16, 17] score a subset by unary scores minus pairwise overlap penalties and are solved heuristically, often by annealing; determinantal selection [18, 19, 20] trades quality against a similarity kernel with greedy MAP inference; facility location [21] assigns Hough votes to hypotheses at a per-hypothesis cost, the closest precedent for the price ρ. Each of them values candidates, so what an instance is worth depends on how many boxes describe it. The objective F of (3) divides each demand unit by its identity mass, so an ideal block counts once however many boxes it holds (Prop. 1a), and its cubic terms (Lemma C2c) place it outside the quadratic family that QUBO solvers represent without auxiliary variables.

Identity partition. Correlation clustering [22] and multicut tracking [23] partition detections into identities under hard transitivity constraints at inference, and [23] clusters redundant detections within frames as well as across them. DIAL-Det leaves the partition implicit: $\mathscr{L}_t$ penalizes, during training, violations of the same triangle inequality written on $1-r$, and inference consumes the resulting masses without solving a partition problem.

Training-side duplicate control. One-to-one set prediction and distinct queries [24, 25, 26, 27, 28, 29] remove most duplicates during training, and one-to-many training with NMS can still outperform one-to-one matching [30]. These methods change the detector. DIAL-Det acts on the exported pool of a frozen detector, and Table 1 shows that one-to-one heads still leave 1.9–3.1% of attainable objects unallocated: less than NMS-based detectors, hence a smaller gain, exactly as the heterogeneity law of Sec. 3 predicts.

Calibration, risk and error analysis. Conformal and risk-controlling detection [32, 33, 34] certifies the coverage of boxes, classes or scores [36]. Our certified risks are the per-image waste and allocation shortfall of the deployed decoding policy, controlled over a finite configuration family with Learn-then-Test [37] and empirical-Bernstein bounds [38] (Theorem C2). Error taxonomies [35] attribute errors after the fact; the decomposition (1) is exact, by maximum matching, and its third term is the one a decoder on a fixed pool controls directly (Lemma C1). Caps change AP on large-vocabulary benchmarks [2]; here the cap is a decision variable, with budgets from $K = 10$ to 200 reported.

What DIAL-Det adds. No prior family combines three properties. (i) The unit of value is the identity: a learned mass makes an ideal block worth one object whatever its number of boxes (Prop. 1a). (ii) The report budget and a price per box enter the decision, so one objective yields an admission rule while slots remain and an exchange rule once the report is full (Prop. 2; Propositions C1–C2); in Table A1 the suppression rules and the quadratic and determinantal selectors apply the cap only after deciding, and facility location prices hypotheses without a cap. (iii) Every image carries a checked optimality gap for the executed objective (Theorem 1, Proposition C3), and the deployed policy, fallback included, carries finite-family risk control (Theorem C2). The detector stays frozen throughout.

Appendix B Evidence model: implementation contract

This appendix specifies the evidence model completely; Table B2 collects every constant in one manifest, and Table B1 fixes the role of every image population.

B.1 Candidate export

Each candidate is a tuple $(\mathrm{id}, b, c, s^{\mathrm{det}}, f)$: source ID, box, class, native score and frozen feature (RoIAlign [39] on the detector's FPN level for two-stage and YOLO models; the decoder query for DETR-type models). The detector's own score floor is applied, at most $N \le 500$ candidates are kept by native score, candidates are never padded, and query–class expansion, ignore masks, preprocessing and evaluator version are recorded. Native score and box are immutable in the *membership track*, where q changes membership only; in the *complete-method track* each method outputs its own scores and DIAL-Det outputs q. For multi-class COCO the global cap K is applied before `cocoeval`, which still applies its own per-category cap.

B.2 Support graph

Overlap edges join class-compatible pairs with IoU $\ge .25$. Rescue edges are the top-4 cosine neighbours of the pre-message appearance feature inside the centre-distance gate

$$D_{ij} = \frac{\|c(b_i) - c(b_j)\|_2}{\max\left(\varepsilon, \frac{1}{2}[\mathrm{diag}(b_i) + \mathrm{diag}(b_j)]\right)} \le 2.0, \tag{B.1}$$

with $c(\cdot)$ the box centre and $\mathrm{diag}(\cdot)$ its diagonal. Both branches are symmetrized by union, deduplicated and augmented with self-edges. A rescue edge is support, not a declaration of shared identity: it only allows the identity head to score the pair. Exact all-pairs similarity costs $O(N^2 d)$ for feature width d; this cost is profiled separately (Table F3).

B.3 Encoder

Features are mapped to 128 channels and concatenated with normalized geometry, native score and a 16-channel class embedding. Two residual graph-attention layers [40] carry four attention heads per branch $b \in \{o, r\}$:

$$m_i^{b,\ell} = \sum_{j \in E_b(i)} \alpha_{ij}^{b,\ell} V^\ell h_j^\ell, \qquad \alpha_{ij}^{b,\ell} = \mathrm{softmax}_{j \in E_b(i)}\, a_b^\top \phi\left(W_b[h_i^\ell \| h_j^\ell \| \psi_{ij}]\right), \tag{B.2}$$

$$g_i^\ell = \sigma\left(v^\top [h_i^\ell \| m_i^{o,\ell} \| m_i^{r,\ell} \| \log(1+|E_o(i)|) \| \log(1+|E_r(i)|)]\right), \qquad h_i^{\ell+1} = h_i^\ell + W^\ell \phi\left(\mathrm{LN}[g_i^\ell m_i^{o,\ell} + (1-g_i^\ell) m_i^{r,\ell}]\right), \tag{B.3}$$

where ψ_{ij} are relative-geometry features, ϕ is a pointwise nonlinearity and the learned gate g_i^ℓ weighs the overlap branch against the rescue branch by the evidence each provides.

B.4 Output heads and calibration

The quality head outputs ten nested probabilities p_{it}, $t \in \{.50, .55, \ldots, .95\}$, with nonnegative logit decrements so that p_{it} is nonincreasing in t, and targets $y_{it} = \mathbf{1}[\max_{g:c_g=c_i} \mathrm{IoU}(b_i, g) \geq t]$. The objective consumes the evaluator's match probability: $q_i = \frac{1}{10}\sum_t p_{it}$ for COCO AP, which averages the same ten thresholds, and $q_i = p_{i,.5}$ for the IoU-.5 evaluators of CrowdHuman (AP_{50}) and CityPersons (MR_{-2}); this is the quantity Prop. 1a requires. The identity head maps the symmetric pair feature $[|h_i - h_j|, h_i \odot h_j, \psi_{ij}^{\mathrm{sym}}]$ to one logit. A positive temperature is fitted on development only [41], and the masses are recomputed after temperature scaling, so the objective consumes calibrated affinities.

B.5 Losses and training

The training loss is

$$\mathscr{L} = \mathscr{L}_q + \mathscr{L}_{\mathrm{pair}} + \beta\,\mathscr{L}_m + \gamma\,\mathscr{L}_t, \qquad \beta = .1,\ \gamma = .05, \tag{B.4}$$

with $\mathscr{L}_q$ the nested quality BCE, $\mathscr{L}_{\mathrm{pair}}$ the edge BCE with importance weights $P(e)/Q(e)$ for the natural support-edge distribution P and the sampling proposal Q (normalization recorded, clipped weights labelled), $\mathscr{L}_m$ the log-mass loss (2) and $\mathscr{L}_t$ the transitivity loss of Sec. 2. $\mathscr{L}_m$ is computed on all support edges, not on the sampled minibatch, because the mass is a sum over the whole neighbourhood. $\mathscr{L}_t$ runs over all support triangles (i, j, k) with $ij, jk \in E$, whether or not $ik \in E$: when $ik \notin E$, $r_{ik} = 0$ and the term penalizes $r_{ij} + r_{jk} > 1$, so a strongly linked chain must close. The training assignment a_i is the highest-IoU admissible class-consistent object (ties by annotation ID); background candidates receive private identities. Optimization uses AdamW with learning rate $3 \cdot 10^{-4}$ and weight decay 10^{-4} for 30 epochs, repeated for five seeds.

B.6 Populations and cross-fitting

Table B1 fixes the role of every population. On CrowdHuman, the deployed R50-FPN trains on all 15,000 training images. Four fold detectors, each trained on 11,250 images, produce out-of-fold candidates on the remaining 3,750, so every training image carries candidates from a detector that never saw it: 13,000 of these images train the encoder and 2,000 form the development split, which carries all selection, including the 24-configuration tuning of every decoder. The encoder therefore trains on exactly the regime it meets at deployment, where the deployed detector processes images it never saw. The 4,370 val images are split once, with a recorded seed, into 2,000 calibration images (risk bounds only) and 2,370 audit images (every reported number); both are processed by the deployed detector, so calibration and deployment share one population. The 8k replicate is a controlled setting, with detector and encoder trained on 8,000 images. Public checkpoints (the COCO detectors, CrowdDet, DDQ, the CrowdHuman DINO-R50 and the CityPersons R50-FPN) may have seen every official training image, so their encoders are cross-fitted: the evaluation split is halved with a recorded seed, each half's encoder is trained, and its decoders tuned, on the other half, and metrics are computed on the union. No risk certificate is claimed for them.

Table B1. Data roles. Every reported CrowdHuman number is measured on the audit split; calibration images enter only the risk bounds of Theorem C2.

Population	Images	Candidates from	Encoder	Used for
CrowdHuman train	15,000	–	–	training the deployed R50-FPN
Out-of-fold, encoder	13,000	fold detector that never saw the image	trained	encoder training
Out-of-fold, development	2,000	fold detector that never saw the image	frozen	all selection: decoder tuning, temperature, (ρ, λ) selection
Calibration (val)	2,000	deployed detector	frozen	risk bounds (9) only
Audit (val)	2,370	deployed detector	frozen	every reported CrowdHuman number
8k replicate	8,000	detector trained on 8,000	trained on 8,000	controlled replicate (Table 1)
Public checkpoints	eval. split	public detector	cross-fitted on halves	metrics on the union; no risk certificate

Table B2. Configuration manifest: every constant that defines the evaluated method.

Stage	Setting
Candidate export	native score floor; at most $N = 500$ candidates by native score; no padding; global cap K applied before evaluation
Support	overlap edges at IoU $\geq .25$ (class-compatible); rescue edges: top-4 cosine neighbours inside gate (B.1) ≤ 2.0; union, deduplication, self-edges
Encoder	128-channel projection + normalized geometry, native score, 16-channel class embedding; 2 residual layers; 4 heads per branch; learned branch gate (B.3)
Heads	quality: 10 nested probabilities, $t \in \{.50, \ldots, .95\}$; identity: pair feature $[\lvert h_i - h_j\rvert, h_i \odot h_j, \psi_{ij}^{\mathrm{sym}}]$; temperature fitted on development
Training	loss (B.4) with $\beta = .1$, $\gamma = .05$; AdamW, learning rate $3 \cdot 10^{-4}$, weight decay 10^{-4}; 30 epochs; five seeds
Objective	$(\rho, \eta, \lambda, p, \theta) = (.08, .2, 1.25, 2, .05)$, chosen once on CrowdHuman development and fixed for all sixteen settings
Search	enumerate components with ≤ 12 candidates; width-64 beam with first-improvement swaps otherwise; split components above 256 candidates along their weakest edges
Bounds	per-cardinality LPs with coverage assignments, McCormick rows and Boolean-quadric clique cuts on cliques of $\{\widetilde{r} \geq .5\}$, separated for up to three rounds; LP solver HiGHS [51], its output never trusted; grid $\delta = 2^{-30}$; directed rounding (Prop. C3)
Gate and fallback	certified iff $g_F \leq 1.5\%$ of $\max(B_\delta + \varepsilon_g, 1)$; larger gap: returned uncertified; numerical-integrity failure: tuned local rule L
Budgets	$K \in \{10, 25, 50, 100, 200\}$; deployed $K = 100$; evidence computed once, the encoder never sees K
Risk	24 pairs $\rho \in \{.02, .04, .06, .08, .12, .16\} \times \lambda \in \{1, 1.25, 1.5, 2\}$; targets $(.05, .10)$; $\alpha = .05$; $n = 2{,}000$; $J = 48$
Statistics	paired crossed image–seed bootstrap, $B = 2{,}000$; Holm within four families; heterogeneity slope interval from 20,000 replicates
Runtime	FP16 evidence on one A100; allocation on one pinned CPU core; 8 ms is a soft target, not a hard deadline

Appendix C Proofs

This appendix proves the decomposition, Propositions 1–2 and Theorem 1 of the paper, and states and proves the matching audit and the risk guarantee. Throughout, $(\rho, \eta, \lambda, p, \theta) = (.08, .2, 1.25, 2, .05)$ when numbers are quoted.

C.1 Decomposition and the cost of a duplicate

Lemma C1 (Decomposition and full-report exchange)**.** *For every report $S \subseteq V$ with $|S| \leq K$:*

(a) $|G| \geq H \geq Q \geq C \geq C_{\mathrm{ev}} \geq 0$, so every term of (1) is nonnegative and (1) holds;

(b) if the report is full and capacity binds, $|S| = K \leq H$, then $Q - C = |S| - C$: the allocation shortfall equals the number of reported boxes outside a maximum matching of S;

(c) under (b), if $C < Q$ then for every reported box $i \in S$ left unmatched by some maximum matching of S there is a candidate $j \in V \setminus S$ with $\nu_\tau\big((S \setminus \{i\}) \cup \{j\}, G\big) = C + 1$.

Proof. (a) Enlarging the candidate set cannot shrink a maximum matching, so $C = \nu_\tau(S, G) \leq \nu_\tau(V, G) = H$; each match uses a distinct reported box, so $C \leq |S| \leq K$; hence $C \leq \min(K, H) = Q \leq H \leq |G|$. The evaluator's score-ordered assignment is a matching of S, so $C_{\mathrm{ev}} \leq C$. Since $(H - K)_+ = H - Q$, the four terms of (1) are the differences $|G| - H$, $H - Q$, $Q - C$ and $C - C_{\mathrm{ev}}$, which are nonnegative and telescope to $|G| - C_{\mathrm{ev}}$. (b) $Q = \min(K, H) = K = |S|$. (c) The subsets of V that can be matched into G form the independent sets of a transversal matroid whose rank function is $\nu_\tau(\cdot, G)$ [42]. The boxes $I \subseteq S$ matched by the maximum matching are independent with $|I| = C < Q \leq H = \nu_\tau(V, G)$, so the augmentation axiom yields $j \in V \setminus I$ with $I \cup \{j\}$ independent; $j \notin S$, because I is a maximum independent subset of S. Since $i \notin I$, the set $S' = (S \setminus \{i\}) \cup \{j\}$ contains $I \cup \{j\}$, so $\nu_\tau(S', G) \geq C + 1$; and $\nu_\tau(S', G) \leq \nu_\tau(S \setminus \{i\}, G) + 1 = C + 1$. □

Part (c) is the precise content of the paper's statement that on a full report every retained duplicate costs exactly one attainable instance: each reported box outside a maximum matching can be exchanged, one for one, for an additional match.

C.2 Identity mass: proof of Proposition 1

Definition C1 (Ideal evidence)**.** V is partitioned into identity blocks, each one object or one background candidate; $s_{ij} = \widetilde{r}_{ij} = 1$ within a block and 0 across blocks; and each candidate is eligible only for its block's object.

Proof of Proposition 1. (a) For u in block B, $d_u = \sum_j s_{ju} = |B|$, so $\mu_u = 1/|B|$ and $D_B = \sum_{u\in B} \mu_u = 1$. Coverage of B under S is $\eta \sum_{u\in B} \mu_u \max_{i\in S\cap B} q_i = \eta \max_{i\in S\cap B} q_i$ (zero if $S\cap B = \emptyset$), and pair penalties act only within blocks. Suppose S holds two candidates of B; let j be one of minimal quality in $S\cap B$ and $i \neq j$ another. Removing j changes the unary term by $-(q_j - \rho)$, leaves the coverage of B unchanged because a candidate of quality at least q_j remains, and relieves the penalty $\lambda \sum_{k\in S\cap B\setminus\{j\}} w_{jk} \geq \lambda w_{ij} = \lambda q_j$. Hence $F(S\setminus\{j\}) - F(S) \geq \rho + (\lambda - 1)q_j \geq \rho > 0$, and $S\setminus\{j\}$ is feasible: every optimum keeps at most one candidate per block, and adding a non-best duplicate lowers F by at least ρ. For a duplicate-free S the three terms of (3) give $F(S) = \sum_{i\in S}(q_i - \rho) + \eta \sum_{i\in S} q_i = (1+\eta)\sum_{i\in S} q_i - \rho|S|$. Under Definition C1 distinct representatives target distinct objects, so $C(S) = \sum_{i\in S} \mathbf{1}[i \text{ matches its object}]$, and linearity of expectation with q_i the match probability gives $\mathbb{E}[C(S)] = \sum_{i\in S} q_i$ without any independence assumption. This is (4), and maximizing it under $|S| \leq K$ maximizes expected matches at price $\rho/(1+\eta)$ per box. (b) With $d_u = |B|e^{e_u}$, $D_B = |B|^{-1}\sum_{u\in B} e^{-e_u} \in [e^{-\varepsilon}, e^{\varepsilon}]$. (c) With q and $\widetilde{r}$ fixed, only the coverage term depends on d, and $\mu_u/\mu_u^\star = d_u^\star/d_u = e^{-e_u}$. Writing $b_u = m_u(S) - m_u(T)$,

$$[F_d(S) - F_d(T)] - [F_{d^\star}(S) - F_{d^\star}(T)] = \eta \sum_u (\mu_u - \mu_u^\star) b_u = \eta \sum_u \mu_u^\star (e^{-e_u} - 1)\, b_u,$$

and $|e^{-e} - 1| \leq e^{|e|} - 1$ for all real e gives (5). Only units whose coverage differs between S and T contribute, which makes the bound local. □

C.3 Structure of the objective

Lemma C2 (Structure)**.** *(a) $F = M + \eta\,\mathrm{FL} - \lambda P$ with M modular, $\mathrm{FL}(S) = \sum_u \mu_u \max_{i\in S} q_i \widetilde{r}_{iu}$ monotone submodular and $P(S) = \sum_{\{i,j\}\subseteq S} w_{ij}$ supermodular; hence F is submodular. (b) $F(\emptyset) = 0$ and F takes negative values, so a multiplicative guarantee for nonnegative objectives [45] does not transfer: an α-approximation of $F + c$ only yields $F(\widehat{S}) \geq \alpha F^\star - (1-\alpha)c$. (c) If candidates i, j, k cover a common unit u with $\widetilde{r}_{iu} = \widetilde{r}_{ju} = \widetilde{r}_{ku} = 1$, the coefficient of $x_i x_j x_k$ in the multilinear polynomial of F is at least $\eta\,\mu_u\, q_{(3)} > 0$, with $q_{(3)}$ the smallest of the three qualities; for three equal-quality candidates of a block B it equals $\eta q D_B$. Hence no quadratic pseudo-Boolean function, and in particular no QUBO without auxiliary variables [16, 17], represents F.*

Proof. (a) For each u, $S \mapsto \max_{i\in S} a_{iu}$ with $a_{iu} \geq 0$ is monotone submodular: the marginal of e is $[a_{eu} - \max_{i\in S} a_{iu}]_+$, nonincreasing in S [44]; nonnegative combinations preserve both properties. The marginal of e in P is $\sum_{j\in S} w_{ej}$, nondecreasing in S because $w \geq 0$. (b) An isolated candidate with $(1+\eta)q_e < \rho$ has $F(\{e\}) < 0$. (c) On binary inputs with $a \geq b \geq c \geq 0$, $\max(ax_i, bx_j, cx_k) = ax_i + (1-x_i)[bx_j + (1-x_j)cx_k]$, whose expansion contains $+c\,x_i x_j x_k$; every unit covered by all three candidates contributes a nonnegative cubic coefficient of this form, units covered by fewer contribute none, and the modular and pair terms are at most quadratic. The multilinear representation of a pseudo-Boolean function is unique [46], so F has degree three. □

C.4 The two evidence failures, worked out

Example C1 (Fragmentation)**.** Candidates a_1, a_2, a_3, a_4 of one person and b of another all have quality .9 except $q_b = .85$, with $r_{a_1a_2} = r_{a_3a_4} = 1$ and no other support. Every $d_u = d_u^\star = 2$ on the a side and $d_b = 1$, so $\mathscr{L}_m = 0$, yet the blocks $\{a_1, a_2\}$ and $\{a_3, a_4\}$ are valued separately: a block of quality q is worth $(1+\eta)q - \rho = 1.2q - .08$, so at $K = 2$ the report $\{a_1, a_3\}$ scores 2.00 against 1.94 for $\{a_1, b\}$, and two fragments of one person outvalue two people. The mass target counts support neighbours only, which is why no mass loss can repair this; the rescue branch and the fragmentation rate (Appendix E) address it.

Example C2 (Bridging)**.** Candidates i, k of distinct objects and j between them have $q_i = q_k = .80$, $q_j = .78$, $r_{ij} = r_{jk} = .75$ and $r_{ik} = 0$, so $s_{ij} = s_{jk} = .5625$. Then $d_i = d_k = 1.5625$ and $d_j = 2.125$, and

$$F(\{j\}) = .70 + .2 \cdot .78\left(\tfrac{2\cdot.5625}{1.5625} + \tfrac{1}{2.125}\right) = .886 \;>\; F(\{i\}) = .72 + .2 \cdot .80\left(\tfrac{1}{1.5625} + \tfrac{.5625}{2.125}\right) = .865 :$$

the bridge draws coverage from both neighbours and wins the only slot. The Fréchet bound enforced by $\mathscr{L}_t$ requires $r_{ik} \geq r_{ij} + r_{jk} - 1 = .5$; satisfying it either raises r_{ik}, merging the blocks, which the pair BCE resists for distinct objects, or lowers the bridge's affinities.

C.5 Admission rule: Proposition 2(a), all branches

Proposition C1 (Complete admission cut-off)**.** *Let a kept box i and a candidate j satisfy $q_i \geq q_j > 0$; let $x = r_{ij}^p$, $z = x\mathbf{1}[x \geq \theta]$ and $a = 1 + \ell_j$, with ℓ_j the outside mass of j; let no other kept box have a retained affinity to j, and let j's other units already be covered by kept boxes at least as well as j would cover them. Then*

$$D_j(x) = q_j - \rho - \lambda q_j z + \eta\,[q_j - q_i z]_+/(a+x) \tag{C.1}$$

is nonincreasing in x with a single downward jump at θ, $D_j(1) \leq -\rho < 0$, and j is admitted exactly when $x < \bar{x}$, where, with $A = \lambda q_j$, $B = \lambda q_j a - (q_j - \rho) + \eta q_i$, $C = (q_j - \rho)a + \eta q_j$, $d_- = q_j - \rho + \eta q_j/(a+\theta)$, $d_+ = D_j(\theta)$ and $t_0 = (1-\rho/q_j)/\lambda$,

$$\bar{x} = \begin{cases} 0, & D_j(0) \leq 0, & \text{(i)} \\ \eta q_j/(\rho - q_j) - a, & D_j(0) > 0 \geq d_-, & \text{(ii)} \\ \theta, & d_- > 0 \geq d_+, & \text{(iii)} \\ t_0, & d_+ > 0,\ t_0 \geq q_j/q_i, & \text{(iv)} \\ 2C/\left(B + \sqrt{B^2 + 4AC}\right), & \textit{otherwise}, & \text{(v)} \end{cases} \tag{C.2}$$

The affinity cut-off is $\bar{r} = \bar{x}^{1/p}$. For every kept quality and outside mass, branches (ii)–(iii) require $\rho/(1+\eta) < q_j \leq \rho/(1-\lambda\theta)$, that is $q_j \in (.0667, .0853]$ at the deployed constants, the band quoted in the paper; with $q_i = .9$ and $a = 1$ it narrows to $(\rho/(1+\eta), .0785] \approx (.067, .079]$.

Proof. Under the covering assumption j adds coverage only to its own unit, whose demand is $\mu_j = 1/d_j = 1/(a+x)$ and whose current coverage is $q_i z$; unit i gains nothing because $q_j z \leq q_i$; and the penalty is $\lambda w_{ij} = \lambda z q_j$. This gives (C.1). For $x < \theta$, $z = 0$ and $D_j' = -\eta q_j/(a+x)^2 \leq 0$. At θ, z jumps from 0 to θ, which adds the penalty $\lambda q_j \theta$ and removes residual coverage, a downward step. For $x \geq \theta$ with positive residual, $D_j' = -\lambda q_j - \eta(q_i a + q_j)/(a+x)^2 < 0$; once the residual vanishes ($x \geq q_j/q_i$), $D_j' = -\lambda q_j < 0$. Since $\lambda \geq 1$ and $q_i \geq q_j$, $D_j(1) = (1-\lambda)q_j - \rho \leq -\rho$. So $\{x : D_j(x) > 0\} = [0, \bar{x})$, and $\bar{x}$ is located by the sign pattern: no admission (i); a root below θ, where $q_j - \rho + \eta q_j/(a+x) = 0$ gives (ii); a zero crossing at the jump (iii); a root where the residual has vanished, $q_j - \rho - \lambda q_j x = 0$, which lies in that region exactly when $t_0 \geq q_j/q_i$ (iv); otherwise the root solves $(q_j - \rho)(a+x) - \lambda q_j x(a+x) + \eta(q_j - q_i x) = 0$, that is $Ax^2 + Bx - C = 0$, which is (7) of the paper. In branch (v) the root is positive, and the rationalized form $2C/(B + \sqrt{B^2 + 4AC})$ avoids cancellation when $B \geq 0$ (use $(-B + \sqrt{B^2 + 4AC})/(2A)$ otherwise). Branches (ii)–(iii) need $D_j(0) > 0$, that is $q_j(1 + \eta/a) > \rho$, hence $q_j > \rho/(1+\eta)$ because $a \geq 1$; and they need $d_+ \leq 0$, hence $q_j(1 - \lambda\theta) \leq \rho$ because the residual term of d_+ is nonnegative. At $q_i = .9$, $a = 1$: $D_j(0) > 0 \iff q_j > \rho/(1+\eta)$; $d_- \leq 0 \iff q_j \leq \rho/(1 + \eta/(1+\theta)) \approx .0672$; and $d_+ \leq 0 \iff q_j \leq .0785$. □

Figure C1a,b draws $\bar{r}$ at $q_i = .9$ and $a = 1$. For $q_j \in [.092, .628]$ the cut-off is branch (iv): the kept box already covers j's unit, and j is admitted exactly when its redundancy penalty $\lambda q_j x$ stays below its net value $q_j - \rho$, a rule that involves neither the kept quality nor the outside mass. Coverage moves the cut-off only in the band of Fig. C1b and, above $q_j = .628$, by at most .014; across $q_i \in \{.5, .7, .9\}$ and $a \in \{1, 2, 3\}$ the cut-off varies by less than .02 for $q_j \geq .1$. Admission is therefore, to that accuracy, a threshold set by the candidate's own quality: at $q_i = .9$ a candidate of quality .85 is admitted below $r = .86$ and one of quality .10 below $r = .40$, so every constant threshold t on r, the form of the local rule, errs on at least one of them, and every $t \in (.40, .86)$, .5 included, on both.

C.6 Exchange rule: Proposition 2(b) in closed form

Proposition C2 (Closed-form exchange boundary)**.** *At $K = 2$, let candidates i, i' have qualities $q_i \geq q_{i'} > \rho$ and affinity $s = r_{ii'}^p$, and let a distinct candidate j have quality $q_j > \rho/(1+\eta)$, with no other affinity, so that $d_i = d_{i'} = 1 + s$ and $d_j = 1$. Then $f(s) = F(\{i, j\}) - F(\{i, i'\})$ is increasing in s with a single upward jump at θ, and the optimum over all reports of size at most two keeps i' for $s < s^\star$ and exchanges it for j for $s > s^\star$, where*

$$s^\star = \begin{cases} 0, & q_j > q_{i'}, & \text{(i)} \\ \dfrac{\eta}{(1+\eta)q_j/q_{i'} - 1} - 1, & q_{i'}\left(1 + \frac{\eta}{1+\theta}\right) < (1+\eta)q_j \leq (1+\eta)q_{i'}, & \text{(ii)} \\ \max(\theta, \sigma), & \textit{otherwise}, & \text{(iii)} \end{cases} \tag{C.3}$$

with σ the positive root of

$$\lambda q_{i'}\sigma^2 + \big[\eta q_i + (1+\eta)q_j + (\lambda-1)q_{i'}\big]\sigma - (1+\eta)(q_{i'} - q_j) = 0 \tag{C.4}$$

if $q_i\sigma < q_{i'}$, and $\sigma = \big(q_{i'} - (1+\eta)q_j\big)/(\lambda q_{i'})$ otherwise. The boundary is $r^\star = (s^\star)^{1/p}$; it is nonincreasing in q_j, and strictly decreasing on branch (ii) and wherever $\sigma > \theta$ on branch (iii).

Proof. For $s < \theta$ no affinity is retained: unit i' is uncovered by i, and $f(s) = (1+\eta)q_j - q_{i'} - \eta q_{i'}/(1+s)$, increasing. For $s \geq \theta$, i covers the demand of i' exactly as i' would, $q_i s/(1+s) + q_i/(1+s) = q_i$, so $F(\{i,j\}) = (1+\eta)(q_i+q_j) - 2\rho$, while $F(\{i,i'\}) = q_i + q_{i'} - 2\rho + \eta\,(q_i + \max(q_i s, q_{i'}))/(1+s) - \lambda q_{i'} s$ decreases in s; so f increases, strictly, on both sides of θ. At θ, f jumps up by $\eta q_i\theta/(1+\theta) + \lambda q_{i'}\theta > 0$ if $q_i\theta < q_{i'}$, and by $\eta q_{i'}/(1+\theta) + \lambda q_{i'}\theta > 0$ otherwise. Hence the exchange set is $(s^\star, 1]$, and $f(1) = (1+\eta)q_j + (\lambda-1)q_{i'} > 0$ guarantees $s^\star < 1$. Branch (i): $f(0) = (1+\eta)(q_j - q_{i'}) > 0$. Branch (ii): $f(0) \leq 0 < f(\theta^-)$, and $f(s) = 0$ below θ solves $1 + s = \eta q_{i'}/((1+\eta)q_j - q_{i'})$. Branch (iii): $f(\theta^-) \leq 0$; if the root σ of the retained expression satisfies $\sigma \leq \theta$, then $f(\theta^+) > 0$ and $s^\star = \theta$; otherwise $s^\star = \sigma$. Multiplying $f(s) = 0$ by $1+s$ in the residual region $q_i s < q_{i'}$ gives (C.4); in the saturated region, $f(s) = (1+\eta)q_j - q_{i'} + \lambda q_{i'} s$ is linear. For fixed s, f increases in q_j with slope $1+\eta$ on every branch, so the exchange set grows with q_j and $s^\star$ is nonincreasing; where $s^\star$ is a zero of f, as on branch (ii) and on branch (iii) with $\sigma > \theta$, it strictly decreases, because f strictly increases in both arguments. Finally, $\{i,j\}$ dominates every other report of size at most two: adding j to $\{i\}$ or to $\{i'\}$ gains $(1+\eta)q_j - \rho > 0$; adding i to $\{j\}$ or to $\emptyset$ gains at least $q_i - \rho > 0$; and $F(\{i,j\}) - F(\{i',j\}) = (q_i - q_{i'})(1+\eta\kappa) \geq 0$, with $\kappa = 1$ for $s \geq \theta$ and $\kappa = 1/(1+s)$ below. The optimum is therefore $\{i,i'\}$ or $\{i,j\}$, as the sign of f decides. □

At $(q_i, q_{i'}) = (.9, .85)$, (C.3) evaluates to $r^\star = .84, .56, .36$ at $q_j = .1, .5, .7$, the values of the paper, and it reproduces the paper's bisection ledger at every grid point to within $5 \cdot 10^{-5}$; branch (ii) occupies only $q_j \in (.843, .85]$. Across $q_j \in [.1, .85]$ the boundary stays within .22 of the expected-match boundary $1 - q_j/q_{i'}$ (Fig. C1c). Figure C1d shows the mechanism: the gain f rises with the affinity, jumps at $\theta^{1/p}$, where i begins to cover the unit of i' and the pair penalty switches on, and changes sign at $r^\star$.

Corollary C1 (No quality-blind keep-below rule is optimal)**.** *Let a rule keep i' iff $r_{ii'} < t(q_i, q_{i'})$, with a threshold that does not depend on q_j. Whenever $r^\star$ is strictly decreasing on an interval of q_j — as on branches (ii) and (iii) with $\sigma > \theta$ — the rule disagrees with the optimum on a set of instances of positive measure.*

Proof. Take $q_j < q'_j$ in that interval, so that $r^\star(q_j) > r^\star(q'_j)$, and any $r \in (r^\star(q'_j), r^\star(q_j))$. The optimum keeps i' at (q_j, r) and exchanges it at (q'_j, r), while the rule decides both instances identically; by continuity the disagreement persists on a neighbourhood. □

C.7 Checked gap: executed arithmetic and Theorem 1

Proposition C3 (Executed arithmetic)**.** *Round every coefficient of (3) — $a_i = q_i - \rho$, $b_{iu} = \eta\mu_u q_i \widetilde{r}_{iu}$ and $c_{ij} = \lambda w_{ij}$ — to the nearest multiple of $\delta = 2^{-30}$; this defines the executed objective F_δ.*

(a) Exact lower side. *Every sum of grid coefficients whose partial sums stay below 2^{23} in magnitude is computed exactly in binary64 [47]. Consequently $F_\delta(S)$ for every S, the enumerated and beam lower curves and the lower max-plus program are exact, and $L_\delta = F_\delta(S)$.*

(b) Directed dual bound *[50]. For* $\max\{c^\top v : Av \leq b,\ Hv = h,\ 0 \leq v \leq 1\}$ *and any multipliers $y \geq 0$ and β,*

$$c^\top v \ \leq\ \Phi(y,\beta) = b^\top y + h^\top\beta + \textstyle\sum_\ell \big[c_\ell - (A^\top y + H^\top\beta)_\ell\big]_+. \tag{C.5}$$

Φ is nondecreasing in $b^\top y + h^\top\beta$ and nonincreasing in every $(A^\top y + H^\top\beta)_\ell$. Evaluating $b^\top y + h^\top\beta$ with $\mathrm{rd}_\uparrow$, every $(A^\top y + H^\top\beta)_\ell$ with $\mathrm{rd}_\downarrow$, and every difference, positive part and the outer sum with $\mathrm{rd}_\uparrow$ therefore returns a machine number $\bar{\Phi} \geq \Phi(y,\beta)$, whatever multipliers the solver returns; negative components of y are set to zero, exactly, before evaluation.

(c) Upper program. *The max-plus program over upper values, evaluated with $\mathrm{rd}_\uparrow$ on every addition, returns an upper bound on its exact value.*

(d) Transfer. $\sup_{|S|\leq K} |F(S) - F_\delta(S)| \leq \varepsilon_g = \frac{\delta}{2}\big(K + N + \binom{K}{2}\big)$.

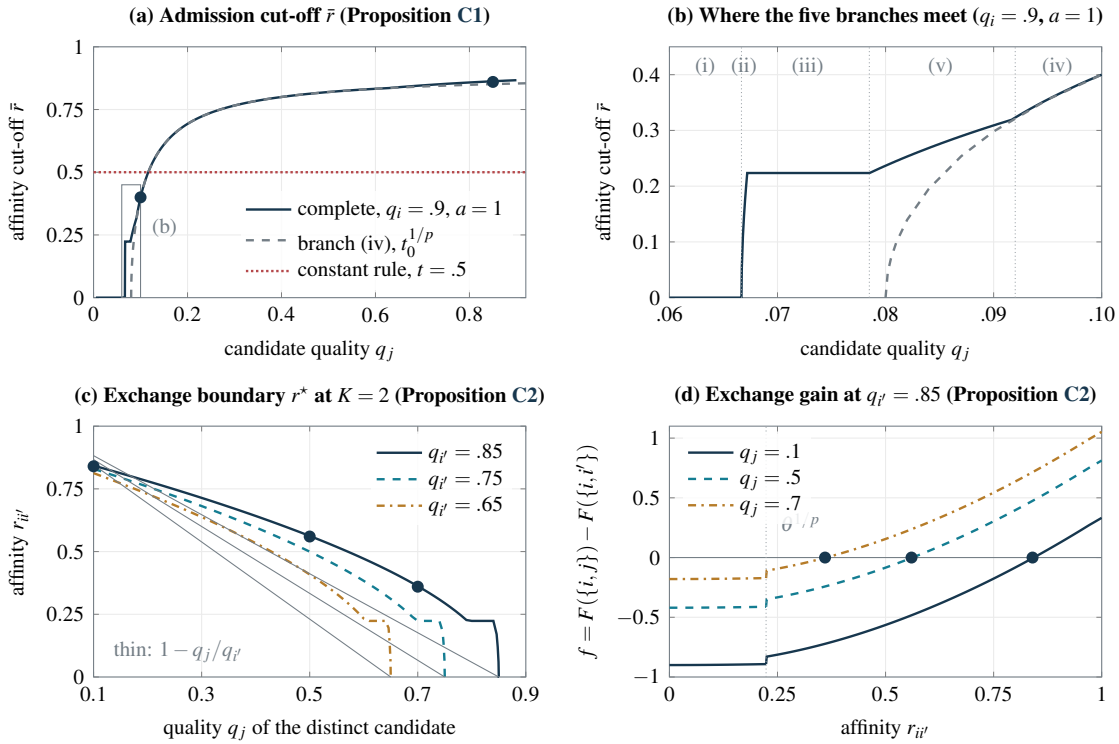


Figure C1. The two derived rules at the deployed constants (analytic, exact). (a) Complete admission cut-off (C.2) for a kept box of quality .9 and $a = 1$; dashed: branch (iv), $t_0^{1/p}$, which the cut-off follows for $q_j \in [.092, .628]$; dots: the paper's worked cases, admitted below $r = .86$ and .40; box: the band magnified in (b). (b) The band where the branches meet: (i) no admission, (ii) the near-vertical rise, (iii) the plateau at $\theta^{1/p} \approx .22$, (v) the quadratic root, (iv) the balance of price and penalty. (c) Closed-form exchange boundary (C.3) for $q_i = .9$: above each curve a full report exchanges the near-duplicate i' for the distinct candidate j; thin lines: expected-match boundaries; dots: the values quoted in the paper. (d) The exchange gain f of Proposition C2: it increases in $r_{ii'}$, jumps up at $\theta^{1/p}$ and changes sign at $r^\star$ (dots), beyond which the exchange is optimal.

Proof. (a) A multiple $m\delta$ with $|m\delta| < 2^{23} = 2^{53}\delta$ has an integer $|m| < 2^{53}$ and is exactly representable; the exact sum of two such numbers is again one of them whenever its magnitude stays below 2^{23}, so round-to-nearest returns it unchanged, and max is exact. Because $q, \mu, \widetilde{r} \in [0,1]$, $|a_i| \le 1$, $b_{iu} \le \eta$ and $c_{ij} \le \lambda$, so every executed magnitude is at most $K + \eta N + \lambda\binom{K}{2} \approx 6.4 \cdot 10^3$ at $K = 100$, $N = 500$, three orders of magnitude below $2^{23} \approx 8.4 \cdot 10^6$. (b) For feasible v, $c^\top v = (A^\top y + H^\top \beta)^\top v + (c - A^\top y - H^\top \beta)^\top v \le y^\top A v + \beta^\top H v + \sum_\ell [c - A^\top y - H^\top \beta]_{\ell,+} v_\ell \le \Phi(y, \beta)$, using $y \ge 0$, $Av \le b$, $Hv = h$ and $0 \le v \le 1$. Since $[c_\ell - t]_+$ is nonincreasing in t, a lower enclosure of t can only increase each term; an upper enclosure of $b^\top y + h^\top \beta$ can only increase Φ; upward-rounded sums of nonnegative terms dominate their exact sums. The bound holds for every $y \ge 0$, so solver inaccuracy loosens it and never invalidates it. (c) Addition and max are nondecreasing in each argument; induction over the stages of the program. (d) Each coefficient moves by at most $\delta/2$. For $|S| \le K$ the objective involves at most K unary terms, at most N coverage terms, one maximum per unit, which moves by at most the largest perturbation, and at most $\binom{K}{2}$ pair terms. □

At $K = 100$ and $N = 500$, $\varepsilon_g = 5550 \cdot 2^{-31} \approx 2.6 \cdot 10^{-6}$, more than three orders of magnitude below the 1.5% gate. The independent checker recomputes (C.5) from the exported multipliers in exact rational arithmetic and accepts the certificate only if the exact value does not exceed the reported $\bar{\Phi}$ (Appendix D).

Proof of Theorem 1. Splitting a component along cut edges removes the coverage terms b_{iu} across the cut and the pair penalties across the cut. For any S, a unit covered across the cut loses at most $\max_{i:(i,u)\text{ cut}} b_{iu}$, and removed penalties only raise the split objective, so $F_\delta(S) \le F_{s,\delta}(S) + A_s$ with $A_s = \sum_u \max_{i:(i,u)\text{ cut}} b_{iu}$. The per-cardinality upper curves bound $F_{s,\delta}$ on each component (Proposition C3b), and the upward-rounded program combines them (Proposition C3c), so $B_\delta \ge \max_{|S| \le K} F_\delta(S)$ with B_δ the upward-rounded sum of that program's value and A_s. The incumbent is re-evaluated on the unsplit grid objective, exactly (Proposition C3a), so $L_\delta = F_\delta(S)$. By Proposition C3d, $F^\star \le \max F_\delta + \varepsilon_g \le B_\delta + \varepsilon_g$ and

$F(S) \geq L_\delta - \varepsilon_g$, which gives (8). □

C.8 The matching audit

Theorem C1 (Matching audit: a bound on allocation misses)**.** *Let S be the report of Algorithm 1 with certified gap g_F (Theorem 1). Build reference blocks from a maximum matching M of the admissibility graph of (V,G): each matched object and its matched candidate seed a block, and every other matchable candidate joins the block of one adjacent matched object. Let F_0 be the objective (3) with the learned q on matchable candidates, $q=0$ on the others, and ideal affinities within blocks (Definition C1). Let $v_1 \geq v_2 \geq \cdots$ be the positive block values of F_0, $Q^+ = \min(K, \#\{b : v_b > 0\})$, $S_0^\star$ an optimum of F_0, $d_0(A) = |F(A) - F_0(A)|$ and $R = d_0(S_0^\star) + g_F + d_0(S)$. Then*

$$Q - C \;\leq\; \min\Big\{Q,\; Q - Q^+ + t^\star\Big\}, \qquad t^\star = \max\Big\{t \in \{0,\dots,Q^+\} : \textstyle\sum_{b=Q^+-t+1}^{Q^+} v_b \leq R\Big\}. \tag{C.6}$$

Proof. M has H edges. An unmatched matchable candidate cannot be adjacent to an unmatched object, since the edge would augment M [43]; attaching it to an adjacent matched object therefore creates no new block, and exactly H blocks exist even when eligibility is ambiguous. Every candidate of a block is eligible for the block's object. By Prop. 1a, block values are $v_B = (1+\eta)\max_B q - \rho$, $F_0^\star = \sum_{b \leq Q^+} v_b$ (so $Q^+ \leq \min(K,H) = Q$), and $F_0(S)$ is at most the sum of the values of the m positive blocks S represents. These blocks have distinct objects, each eligible for a representative in S, so $m \leq C$ and $F_0(S) \leq \sum_{b \leq m} v_b$. With $t = Q^+ - m$, $F_0^\star - F_0(S) \geq \sum_{b=Q^+-t+1}^{Q^+} v_b$. On the other side, $F_0^\star = F_0(S_0^\star) \leq F(S_0^\star) + d_0(S_0^\star) \leq F^\star + d_0(S_0^\star)$ and $F_0(S) \geq F(S) - d_0(S)$, so $F_0^\star - F_0(S) \leq d_0(S_0^\star) + [F^\star - F(S)] + d_0(S) \leq R$. Hence $t \leq t^\star$, $C \geq m \geq Q^+ - t^\star$, and $Q - C \leq Q - Q^+ + t^\star$; trivially $Q - C \leq Q$. □

The bound is computable image by image: $S_0^\star$ is closed form by Prop. 1a, and d_0 compares two objectives on given sets. It requires labels, so it is an audit, not an AP guarantee; for certified integers, R is enclosed upward and the tail sums downward. The tail inversion is tighter than dividing R by v_{Q^+} and isolates the price-induced loss $Q - Q^+$: with $Q = 5$, $Q^+ = 4$, $v = (.92, .68, .41, .15)$ and $R = .50$, only $t = 1$ is admissible and the bound is 2 misses, where division by v_{Q^+} gives 4 (analytic values). We call the bound *non-vacuous* on an image when it is smaller than Q; for fallback outputs no g_F exists and the bound is Q.

C.9 Risk control of the deployed policy

Theorem C2 (Finite-family risk control)**.** *Let Λ be the 24 configurations (ρ, λ) of Table B2. For $c \in \Lambda$ and $k \in \{w, a\}$, let $R_k(c) \in [0,1]$ be the per-image waste or shortfall of the complete deployed policy — Algorithm 1 with its uncertified and fallback branches — at configuration c, and let the n calibration images be i.i.d. from the deployment distribution. Let $U_k(c)$ be the bound (9) with $h = \log(2J/\alpha)$ and $J = 2|\Lambda| = 48$. Then, with probability at least $1-\alpha$, $\mathbb{E}[R_k(c)] \leq U_k(c)$ for all $c \in \Lambda$ and k simultaneously. On that event every configuration selected from $\widehat{\Lambda} = \{c : U_w(c) \leq .05,\ U_a(c) \leq .10\}$, by any rule — in particular the development-AP maximizer — meets both targets.*

Proof. For i.i.d. losses in $[0,1]$ with sample mean $\bar{R}$ and unbiased sample variance s^2, the empirical-Bernstein inequality [38, Thm. 4] gives $\mathbb{E}R \leq \bar{R} + \sqrt{2s^2 \log(2/\delta')/n} + 7\log(2/\delta')/(3(n-1))$ with probability at least $1-\delta'$. Setting $\delta' = \alpha/J$ and taking a union bound over the J hypotheses gives the simultaneous statement with $h = \log(2J/\alpha)$. On that event every $c \in \widehat{\Lambda}$ has true risks at most its bounds, hence at most the targets, and the selected configuration belongs to $\widehat{\Lambda}$ [37, 36]. □

The losses $R_w = (|S| - C)/K$ and $R_a = (Q - C)/Q$ (0 if $Q = 0$) lie in $[0,1]$ by Lemma C1a. Exchangeability holds by construction: calibration and audit are a seeded random split of the val images, processed by one deployed detector that saw neither (Table B1). With $J = 48$ and $\alpha = .05$, $h = \log 1920 \approx 7.56$. The high-pressure stratum, 400 calibration images, carries the same bound but no target: at that n the additive term $7h/(3(n-1)) \approx .044$ alone rules out certifying a useful target.

Appendix D Solver, certificate and failure policy

D.1 Exact program

With $a_i = q_i - \rho$, $b_{iu} = \eta\mu_u q_i \widetilde{r}_{iu}$ and $c_{ij} = \lambda w_{ij}$, maximizing (3) under $|S| \le K$ is the mixed-integer program

$$\max \sum_i a_i x_i + \sum_{i,u} b_{iu} y_{iu} - \sum_{i<j} c_{ij} z_{ij} \quad \text{s.t.} \quad \begin{array}{l} x \in \{0,1\}^N,\ \sum_i x_i \le K,\ 0 \le y_{iu} \le x_i,\ \sum_i y_{iu} \le 1, \\ z_{ij} \ge x_i + x_j - 1,\ 0 \le z_{ij} \le \min(x_i, x_j). \end{array} \tag{D.1}$$

For binary x the coverage simplex $\sum_i y_{iu} \le 1$, $y_{iu} \le x_i$ selects the best selected representative of each unit, so $\sum_i b_{iu} y_{iu} = \max_{i \in S} b_{iu}$ at the optimum, and the McCormick rows [48] force $z_{ij} = x_i x_j$ because the objective penalizes z. The program contains no set-packing rows $x_i + x_j \le 1$: these would forbid two distinct objects whose boxes overlap, which is exactly the case the decoder must be able to report.

D.2 Relaxation and clique cuts

Relaxing $x \in [0,1]^N$ gives an upper bound, which is weakest on duplicate cliques: at $x \equiv \frac{1}{2}$ the McCormick rows allow $z_{ij} = 0$, so a half-selected clique pays no penalty. Boolean-quadric clique inequalities [49] repair this.

Lemma D1 (Clique cuts)**.** *For every set Q' of candidates and integer $k' \ge 0$, $\sum_{\{i,j\} \subseteq Q'} z_{ij} \ge k' \sum_{i \in Q'} x_i - k'(k'+1)/2$ holds for all binary x with $z_{ij} = x_i x_j$. On a clique of size c at $x \equiv \frac{1}{2}$, the cut with $k' = \lfloor (c-1)/2 \rfloor$ forces $\sum z_{ij} \ge c(c-2)/8$ for even c and $\ge (c-1)^2/8$ for odd c, where the McCormick rows force nothing.*

Proof. With $t = \sum_{i \in Q'} x_i$, the left side is $\binom{t}{2}$, and $\binom{t}{2} - k't + k'(k'+1)/2 = (t-k')(t-k'-1)/2 \ge 0$ for integer t. At $x \equiv \frac{1}{2}$ the right side is $k'(c-k'-1)/2$, which equals the stated values at $k' = \lfloor (c-1)/2 \rfloor$. □

Cuts are separated at the LP optimum: for each maximal clique of $\{\widetilde{r} \ge .5\}$ with $\sum_{Q'} x > 1$, the inequality with $k' = \lfloor \sum_{Q'} x \rfloor$ — the most violated one — is added when violated, for up to three rounds.

D.3 Cardinality curves and max-plus combination

For a component $\mathscr{C}$ and each exact cardinality $k = 0, \dots, \min(|\mathscr{C}|, K)$, the lower curve $\ell_{\mathscr{C}}(k)$ is the grid value of the best set of size k found (enumeration or beam), and the upper curve $u_{\mathscr{C}}(k)$ is the directed dual bound (C.5) of the relaxation with $\sum_i x_i = k$ (for enumerated components $u_{\mathscr{C}} = \ell_{\mathscr{C}}$, exactly). Infeasible sizes carry $-\infty$, which is distinct from a negative value, and $k = 0$ carries the value 0 of the empty set. Components are combined by

$$\mathscr{L}_t(b) = \max_{0 \le k \le \min(b, |\mathscr{C}_t|)} \big[\mathscr{L}_{t-1}(b-k) + \ell_{\mathscr{C}_t}(k) \big], \qquad \mathscr{U}_t(b) = \max_{0 \le k \le \min(b, |\mathscr{C}_t|)} \mathrm{rd}_\uparrow \big[\mathscr{U}_{t-1}(b-k) + u_{\mathscr{C}_t}(k) \big], \tag{D.2}$$

from $\mathscr{L}_0(0) = \mathscr{U}_0(0) = 0$ and $-\infty$ elsewhere; the lower program is exact on the grid (Proposition C3a), and backtracking its maximizer over $b \le K$ returns S.

Remark D1. A Lagrangian relaxation of the cardinality row gives $u_{\mathscr{C}}(k) \le \bar{g}(\vartheta) + \vartheta k$ for every multiplier ϑ, where $\bar{g}(\vartheta)$ is the directed dual bound with the row priced out. Any finite set of multipliers therefore bounds the whole curve with a few LPs; this trades tightness for time without affecting validity.

D.4 Search

Components with at most 12 candidates are enumerated. Larger components are searched by a width-64 beam over insertions: level $k+1$ expands every set of level k by each outside candidate e, scored by its exact grid marginal (6), and keeps the 64 best distinct sets, ties broken by sorted candidate IDs. For every size the best set is then refined by first-improvement single swaps, in candidate-ID order, until no swap improves F_δ; this terminates because F_δ strictly increases over a finite set. The lower curve stores the best set of every size.

D.5 Certificate record and independent checker

Every decoded image exports the record of Table D1. The checker re-reads it without access to the encoder weights, recomputes $L_\delta = F_\delta(S)$ and every upper value (C.5) in exact rational arithmetic from the exported multipliers, re-runs both programs (D.2) and A_s, and accepts the certificate only if the exact recomputations do not exceed the reported upper values and reproduce L_δ exactly.

Table D1. Certificate record exported for every image.

Field	Content
Identity	image ID, candidate-export hash, configuration hash, K, $(\rho, \eta, \lambda, p, \theta)$
Instance	grid coefficients a, b, c as integers in units of δ; component partition; split edges and A_s
Lower side	per component and size: best set (candidate IDs) and exact value; lower program table and its argmax
Upper side	per component and size: LP data hash, multipliers (y, β) and $\bar{\Phi}$; upper program table
Outcome	S (candidate IDs), L_δ, B_δ, ε_g, g_F, status, per-stage timing

D.6 Status and failure policy

Every image receives one status. *Rounded optimum*: $B_\delta = L_\delta$, so $F^\star - F(S) \le 2\varepsilon_g$. *Certified*: $g_F \le 1.5\%$ of $\max(B_\delta + \varepsilon_g, 1)$; the floor 1 keeps near-empty images, where a relative gap is ill-conditioned, from failing the gate. *Uncertified*: a larger gap; S is still returned. *Fallback*: a numerical-integrity failure — a non-finite coefficient, a grid partial sum reaching 2^{23}, an LP without returned multipliers, or a checker rejection — returns the tuned local rule L. Uncertified and fallback outputs are part of the deployed policy: they enter every accuracy, latency and risk figure, and their shares are reported (Table F2). No AP is credited to replaying a set through the checker, and no hard deadline is claimed.

D.7 Verification suite

A script enumerates every report on random instances of up to twelve candidates and checks objective values, marginals (6), exchanges $X_{i\to j}$, the certificates of Theorem 1, the closed forms of Propositions C1–C2 against bisection, and Examples C1–C2. The tests corroborate the proofs above; they do not replace them.

Algorithm D1 Checked allocation, complete form of Algorithm 1

Require: candidates V with (q, r, d) computed once (the encoder never sees K); budget K; constants $(\rho, \eta, \lambda, p, \theta)$
Ensure: IDs of S with native scores; $F(S)$; gap g_F; status; certificate record (Table D1)
$s \leftarrow r^p$; $\tilde{r} \leftarrow s\,\mathbf{1}[s \ge \theta]$; $\mu_u \leftarrow 1/d_u$; round a, b, c to the grid δ ▷ executed objective F_δ
split the $\tilde{r}$-graph into components; split every component above 256 candidates along its weakest edges; record A_s
for each component $\mathscr{C}$ **do**
  if $|\mathscr{C}| \le 12$ **then** enumerate all subsets: $\ell_{\mathscr{C}}(k) = u_{\mathscr{C}}(k)$ for $k \le \min(|\mathscr{C}|, K)$
  else beam of width 64 with first-improvement swaps $\to \ell_{\mathscr{C}}$; per-cardinality LPs with clique cuts, directed dual bounds (C.5) $\to u_{\mathscr{C}}$
  end if
end for
lower program (D.2), exact $\to S$; upper program (D.2), upward-rounded; $B_\delta \leftarrow \mathrm{rd}_\uparrow[\max_{b \le K} \mathscr{U}_m(b) + A_s]$
$L_\delta \leftarrow F_\delta(S)$ on the unsplit grid objective; $g_F \leftarrow \mathrm{rd}_\uparrow[B_\delta - L_\delta + 2\varepsilon_g]$
if a numerical-integrity check fails **then return** the tuned local rule's report, status *fallback*
end if
status $\leftarrow$ *rounded optimum* if $B_\delta = L_\delta$; else *certified* if $g_F \le 1.5\% \cdot \max(B_\delta + \varepsilon_g, 1)$; else *uncertified*
return S with native scores, $F(S)$, g_F, status; export the certificate record

Appendix E Measurement protocol

E.1 Definitions, populations and weighting

Every reported quantity is computed from one per-image record holding K, H, Q, $|S|$, C, C_{ev}, method, seed, population, matching convention and status. Table E1 fixes each quantity's formula, population and weighting. *Pooled* means a ratio of sums over images (object-weighted); *image-weighted* means a mean of per-image values. The two differ, and each quantity uses exactly one of them.

Table E1. Reported quantities: definition, population and weighting.

Quantity	Definition	Population, weighting	Reported
r_N, r_L, r_S	$\sum_i (Q_i - C_i)/\sum_i Q_i$ for native, L, S at $K = 100$	setting's evaluation images; pooled	Table 1, Fig. 3d
Recovered rate	$r_L - r_S$	same; pooled	Fig. 3d
Δ_{100}, Δ_{10}	dataset metric of S minus that of L (CityPersons sign-flipped)	five-seed mean of the dataset metric	Table 1
R_w, R_a	$(\lvert S\rvert - C)/K$; $(Q-C)/Q$, 0 if $Q = 0$	calibration or audit; image-weighted	(9), Table F1
Full share; binding share	$\Pr(\lvert S\rvert = K)$; $\Pr(H > K)$	audit images; image-weighted	Sec. 3, Table F5
Admission, replacement	recovered matches by type (Sec. E.7), as % of $\sum_i Q_i$	audit; pooled	Fig. 5a
Refill rate	slots freed by a removed duplicate and refilled by a matchable candidate, over freed slots	audit; pooled over slots	Table F5
Pair recall	same-object candidate pairs joined by a support edge; object-balanced variant: mean within objects with ≥ 2 eligible candidates	audit same-object pairs; pooled	Table F4
Fragmentation rate	objects with ≥ 2 eligible candidates spread over ≥ 2 components of the $\tilde{r}$-graph	audit objects with ≥ 2 eligible candidates	Sec. 3
Bridging rate	selected boxes with $\tilde{r} > 0$ to eligible candidates of two distinct objects	audit selected boxes	Sec. 3
Merge, split rate	distinct-object pairs with $r > .5$; same-object pairs with $r < \theta^{1/p}$	eligible support pairs of that kind	Sec. 3, Table F4
Log-mass error	quantiles of $\lvert e_u\rvert$	audit demand units	Sec. 3
Mass-robust share	search exchanges whose reference margin exceeds the right side of (5)	audit; pooled over exchanges	Sec. 3
Status shares	rounded optimum, certified, uncertified, fallback	audit images	Sec. 3, Table F2
Audit non-vacuity	Pr(bound (C.6) $< Q$)	audit images with $Q > 0$	Sec. 3
Decode time	wall-clock per image, all stages	audit images; mean, p99	Sec. 3, Table F3

E.2 Matched controls

The local rule L keeps candidates in decreasing q whose affinity r to every kept box is below a threshold tuned on development with the 24-configuration budget. The factorial crosses the objective — Q, the quadratic objective ($\eta = 0$) with its weights retuned, or F, the full objective (3) — with the search — G, repeatedly adding the best positive marginal (6), or S, Algorithm 1 — and with the mass loss, $\beta = 0$ or $\beta = .1$, the encoder being retrained in each case. All eight cells share features, support, training data and tuning budget. The factorial's tests are the three contrasts at the full cell, each factor switched with the other two at their full-model levels, and the interaction $(\mathrm{F,S,on} - \mathrm{F,S,off}) - (\mathrm{Q,S,on} - \mathrm{Q,S,off})$. The pair-derived rule of Proposition C1 and fixed-order marginals are reported in the repository.

E.3 Removals

R: no rescue branch. M: $\beta = 0$, identical to the mass-loss contrast at the full cell and counted once. T: $\gamma = 0$. N: $\mu \equiv 1$ with η retuned on development. P: $\rho = 0$. A: appearance features removed (geometry, score and class only). Each removal is retrained or retuned under the same budget as the full model.

E.4 External decoders and the two tracks

All decoders run on the frozen exports with an equal budget of 24 development configurations (Table E2), each with its documented evidence. In the *membership track* a method chooses membership only and the frozen native scores are output; for re-scoring methods (Soft-NMS [3], CP-Cluster [62], IoU-aware calibration [31]) the re-scored ranking selects the top-K members and the native scores are then restored. The *complete-method track* keeps the same members with each method's own output scores, DIAL-Det outputting q; this track carries no frozen-score certificate. Adaptations marked * replace the method's similarity by r and its unary by q. DPP-MAP* symmetrizes r, projects it onto the PSD cone, adds a

ridge and normalizes the diagonal to one (R_+), sets $L_{\mathrm{DPP}} = \zeta\,\mathrm{diag}(q+\varepsilon)R_+\mathrm{diag}(q+\varepsilon)$ with $\zeta > 0$ tuned, and maximizes $\log\det L_{\mathrm{DPP},S}$ greedily under $|S| \le K$; without ζ the empty set would be optimal, because $\det L_{\mathrm{DPP},S} \le \prod_{i\in S} q_i^2 \le 1$. Methods that need extra detector heads (density or visible region [10, 11, 12, 13, 14]) are retrained-system comparisons and are not reported as frozen-pool baselines.

Table E2. Development grids, 24 configurations per decoder; the grid is fixed before tuning and released with the configuration.

Decoder	Evidence	Grid
Local rule L	our (q,r)	threshold $t \in \{.300, .325, \ldots, .875\}$ (24)
Greedy NMS	native	IoU threshold $\in \{.30, .35, \ldots, .85\}$ (12) $\times$ score floor $\in \{10^{-3}, .05\}$ (2)
Soft-NMS [3]	native	Gaussian $\sigma \in \{.1, .2, \ldots, .8\}$ (8) $\times$ score floor $\in \{10^{-3}, .01, .05\}$ (3)
DIoU-NMS [4]	native	IoU threshold $\in \{.30, \ldots, .85\}$ (12) $\times$ distance exponent $\in \{.6, 1.0\}$ (2)
CP-Cluster [62], ASA-NMS [9], ASA-NMS*	documented	published defaults at the centre of a 6×4 grid over their two documented parameters
IoU-aware calib. [31], Seq2Seq-NMS [8]	learned	calibrator or network trained on the encoder's training split; decision threshold, 24 values
App. QUBO [17]	appearance	unary, overlap and appearance weights, $4 \times 3 \times 2$
DPP-MAP* [20]	our (q,r)	ζ, 8 log-spaced values $\times$ ridge $\in \{10^{-4}, 10^{-3}, 10^{-2}\}$ (3)

E.5 Detector specifics

YOLO26s: one checkpoint, both heads evaluated separately with fixed exports, O2M with its NMS as native and O2O without. CrowdDet: proposal and set IDs are retained so that native Set-NMS stays faithful. DDQ: DIAL-Det re-selects DDQ's post-decoder output, after its internal query NMS; replacing that NMS changes downstream computation and is a separate architecture experiment.

E.6 Identity-evidence tests

Oracle sequence. With geometry and scores fixed: learned (q,r); true evaluator-matched quality; identity oracle restricted to the learned support; unrestricted identity oracle; both oracles; both with exhaustive search.

Corruption test. On audit images, a stated share of same-identity support links is removed at random (fragmentation), or the same share of false links is added between eligible candidates of distinct objects that overlap by IoU $\ge .1$ (bridging); masses are recomputed, the pool is decoded with the deployed model and with $\gamma = 0$, and $Q - C$ is reported.

Dense-group test. Maximal cliques of b persons whose boxes pairwise overlap by IoU $\ge .3$ are taken from the audit annotations; the pool is restricted to candidates eligible for those persons and their duplicates, $K = b$, every method decodes under its tuned configuration, and the share of the b persons matched is averaged over groups.

Clone test (appendix only). The best candidate of one identity per image is duplicated $b-1$ times with identical box and score, so that H, Q and the evaluator are unchanged; the pool is re-encoded and decoded. At $b = 8$ the measured $\Delta\mathrm{AP}_{50}$ is $-.97$ for unnormalized coverage, $-.15$ without the mass loss and $-.06$ for DIAL-Det (Fig. F2b).

E.7 Budgets and the decomposition of gains

Evidence is computed once and reused for $K \in \{10, 25, 50, 100, 200\}$; the encoder has no K input. The shares of $H > K$ and of $|S| = K$ are reported for each K. For each image the maximum matchings of L and S are compared: a recovered match is an *admission* if S matches an object with a candidate that L did not report while L's report was below K, and a *replacement* if L's report was full and S dropped, for that candidate, one matched to no object or a second candidate of a matched object. *Removals* are unmatched outputs dropped without a replacement, reported per image. The refill rate is the fraction of slots freed by removing a duplicate that the optimum refills with a matchable candidate, verified by maximum rematching.

E.8 Heterogeneity and the pre-registered prediction

Per setting, r_N, r_L and r_S are pooled shortfall rates at $K = 100$ (Table E1); the recovered rate $r_L - r_S$ is regressed on r_N by least squares through the origin, with a 95% bootstrap interval over settings (20,000 replicates) and Pearson's r. The HP subset shares images with the full Faster R-CNN COCO setting; without it the slope is .14 ($r = .92$). The focal detector's recovered rate is predicted from the line fitted on the other fifteen settings (slope .15), .85 against .87 observed; the prediction was registered before the focal audit was run. The image-weighted means of Table F1 differ from these pooled rates by construction.

E.9 Statistics

The estimand is the mean dataset metric over five encoder seeds with the detector fixed. Intervals come from a paired crossed image–seed bootstrap [52]: each of $B = 2{,}000$ replicates resamples images and seeds independently with replacement, recomputes the dataset metric of both decoders on the same resampled indices, and forms their difference; intervals use the normal approximation, and the half-width of a paired contrast is $w = 1.96\sqrt{\sigma^2_{\text{img}}/n + \sigma^2_{\text{seed}}/5}$. Holm's procedure [53] runs within the four families of Table E3, fixed before the audit; one additional predeclared test pools the five one-to-one COCO contrasts, their mean with half-width $\sqrt{\sum_i w_i^2}/5$.

Table E3. Test families, fixed before any audit measurement; Holm at $\alpha = .05$ within each family.

Family	Tests	Members
Allocation	16	$S - L$ at $K = 100$, one per setting of Table 1
Budget	16	$S - L$ at $K = 10$, one per setting
Mechanism	9	objective, search and mass-loss contrasts and their interaction (factorial); removals R, T, N, P, A
External	11	DIAL-Det against each re-run decoder of Table 2
Pooled one-to-one	1	mean of the five one-to-one COCO contrasts

E.10 Risk

The deployed encoder and K are fixed before calibration. The 24 pairs of Table B2 are tested against per-image waste and shortfall on the 2,000 calibration images, with $J = 48$, $h = \log(2J/\alpha)$, $\alpha = .05$ and targets $(.05, .10)$ fixed in advance, running the complete policy of Algorithm 1 including uncertified and fallback outputs; the development-AP maximizer among feasible pairs is selected (Theorem C2). HP, the top quintile of images by the attainable-object count H (400 calibration and 474 audit images), is reported with the same bound but carries no target, because its sample cannot certify one below about .15; it serves auditing only, never routing at test time. The audit reports n, $\bar{R}$, s^2, U and the realized audit mean for each risk (Table F1), and the number of feasible configurations.

E.11 Runtime

Timing is measured per single image, with FP16 evidence on one A100 and allocation on one pinned CPU core. Support, encoder, transfer, allocation and checking are reported separately with their mean, the p99 and maximum of the total, the share above 8 ms, edge counts and component statistics; feature access reuses the detector's RoIAlign or query features and is not counted twice. The 8 ms figure is a soft target, not a hard real-time guarantee.

Appendix F Extended results

All results are on the audit split of the focal CrowdHuman R50-FPN at $K = 100$ unless stated.

F.1 Per-setting gains

The sixteen settings pair COCO [63] with YOLOv8n [54], YOLOv12-L [55], both heads of YOLO26s [29], Faster R-CNN [56] and its HP subset, D-FINE-L [57], DEIM-L [58], DEIMv2-S [59] and DINO-R50 [60]; CrowdHuman [64] with the focal R50-FPN [66], its 8k replicate, DINO-R50, CrowdDet [61] and DDQ-DETR [26]; and CityPersons [65]

with an R50-FPN. Figure F1 redraws Table 1 as intervals. Every point estimate is positive at both budgets, and the gain at $K = 10$ exceeds the gain at $K = 100$ in every setting, as the exchange rule predicts once slots bind. The settings whose $K = 100$ gain survives Holm, 7 of 16, are the NMS-based COCO and CrowdHuman settings, which leave more to allocate than every other detector on their dataset; CityPersons leaves a comparable share, but on the smallest evaluation set its interval is too wide to survive the correction.

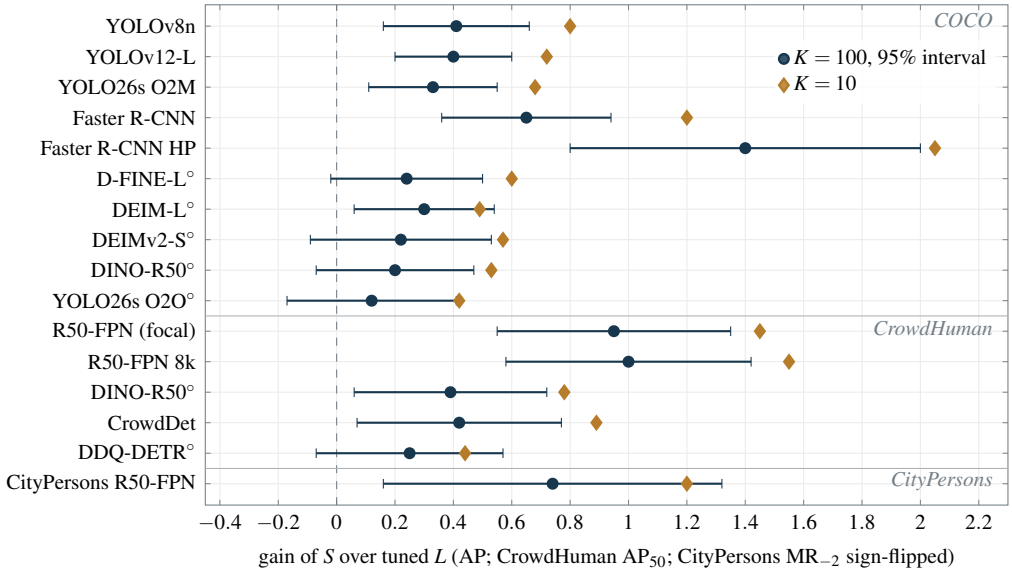


Figure F1. Per-setting gains of Table 1: $K = 100$ with 95% paired intervals (circles) and $K = 10$ (diamonds); °: one-to-one detector.

F.2 Risk

Table F1 reports the calibration statistics behind Theorem C2 and the realized audit means; Figure F2a shows the bounds of all 24 configurations. Of the 24 configurations, 17 are feasible, and the deployed $(\rho, \lambda) = (.08, 1.25)$ is the development-AP maximizer among them. Waste falls and shortfall rises along the price axis, so the price is the knob that trades the two risks, and the redundancy weight shifts the whole frontier.

Table F1. Risk audit, complete policy: calibration statistics, bound (9) ($J = 48$, $\alpha = .05$), predeclared target and audit mean. HP is reported, not certified.

Risk	n	$\bar{R}$	s^2	U	target	audit
Waste, all	2,000	.023	.0042	.037	.05	.023
Shortfall, all	2,000	.048	.0105	.066	.10	.046
Shortfall, HP	400	.080	.0115	.145	–	.078

F.3 Certificate

Table F2 splits the audit by status. The gate withholds certification from few images, and where the optimum is known the returned set is optimal far more often than the certificate can prove: a failed gate marks missing evidence, not a bad allocation. The matching audit of Theorem C1 is non-vacuous on most images, and its average bound sits well above the realized misses, as a worst-case inversion must.

F.4 Workload and latency

Table F3 reports the workload that sets the allocator's cost and the latency of every stage; Figure F3b shows how latency grows with the candidate count. Most candidates fall in components small enough to enumerate, so the cost is governed by the tail of the largest component rather than by N itself, and the p99 overrun of the soft target comes from images at the candidate cap.

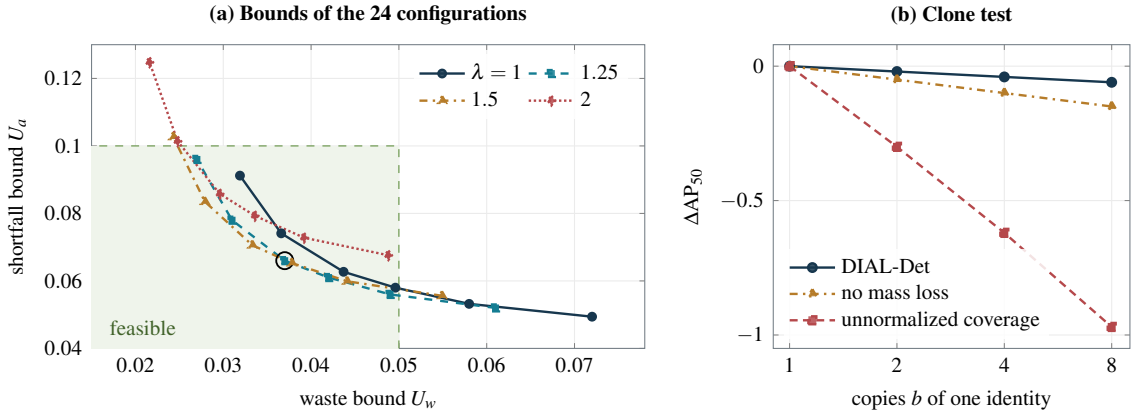


Figure F2. (a) Empirical-Bernstein bounds (9) of the 24 configurations; ρ increases along each series from lower right to upper left; shaded: both targets met; circled: the deployed $(\rho, \lambda) = (.08, 1.25)$. (b) Clone test of Appendix E: H, Q and the evaluator are unchanged by construction, so every drop is allocation error; normalization makes the report nearly blind to multiplicity.

Table F2. Certificate statistics on the audit split.

Quantity	Population	Value
Rounded optimum, $B_\delta = L_\delta$	audit images	78.0%
Certified within the gate, $B_\delta > L_\delta$	audit images	20.8%
Uncertified, S returned	audit images	1.16%
Fallback to L	audit images	.04%
Gate ratio $g_F / \max(B_\delta + \varepsilon_g, 1)$: p50 / p90 / p99	images not rounded optima	.21 / .86 / 2.4%
Exact optimum of F_δ obtained off line by (D.1)	audit images	99.6%
Certified gap vs. true gap, median	certified images	.19 vs. 0%
Returned set optimal	uncertified images	61%
Miss bound (C.6) non-vacuous	audit images with $Q > 0$	91.8%
Miss bound vs. realized misses, mean per image	audit images	2.9 vs. .71

Table F3. Workload and latency, focal setting, audit split.

Quantity	Population	Value
Candidates N: mean / p99	audit images	312 / 500
Support edges, overlap / rescue: mean	audit images	1,620 / 1,010
Retained edges ($\widetilde{r} > 0$) per candidate	audit candidates	1.9
Components of the $\widetilde{r}$-graph per image: mean	audit images	118
Component size, candidate-weighted median	audit candidates	5
Largest component: median / p99	audit images	22 / 181
Enumerated (≤ 12 candidates): components / candidates	audit images	97.6 / 71%
Images with a component split (> 256)	audit images	.4%
LPs per image: median / p99	audit images	6 / 64
Support / encoder / transfer (ms, mean)	audit images	.21 / 1.50 / .07
Allocation / checking (ms, mean)	audit images	1.55 / .32
Total (ms): mean / p99 / max	audit images	3.65 / 8.40 / 23.8
Share above the 8 ms soft target	audit images	1.3%

F.5 Identity evidence

Table F4 extends the identity-evidence paragraph of Sec. 3 to every diagnostic and every evidence removal. Each component moves the failure it was designed against — rescue edges fragmentation, the transitivity loss bridging and false merges, the mass loss the log-mass error and the share of mass-robust exchanges — and leaves the others essentially unchanged. Figure F3a shows the whole distribution of the log-mass error.

Table F4. Identity-evidence diagnostics (%, log-mass error in nats), full model and evidence removals; definitions in Table E1.

Model	pair recall	fragment.	bridging	merges	splits	$\lvert e_u \rvert$ p50 / p90	mass-robust
Full model	97.8	.6	.6	2.0	1.7	.09 / .28	96.2
−R, no rescue branch	94.1	1.9	.6	2.0	1.8	.10 / .30	95.7
−T, $\gamma = 0$	97.8	.6	1.4	3.1	1.6	.09 / .29	95.9
−M, $\beta = 0$	97.7	.7	.7	2.1	1.7	.15 / .40	88.0

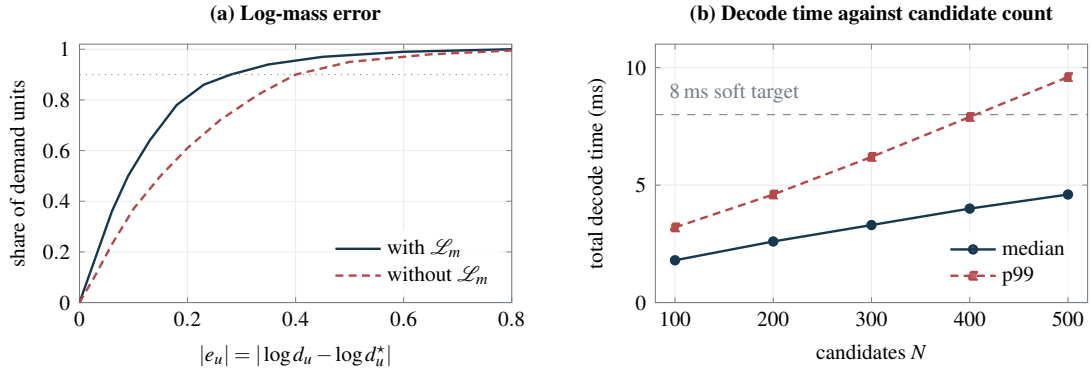


Figure F3. (a) Distribution of the log-mass error over audit demand units with and without the mass loss; the dotted line marks the 90th percentile reported in Sec. 3. (b) Median and p99 of the total decode time per candidate-count bin.

F.6 Budgets

Table F5 completes Fig. 5 of the paper. As K tightens, the share of binding images and of full reports rises together, the source of recovered matches moves from admissions to replacements, and the gain grows in both settings: the admission rule and the mass decide the loose regime, the exchange rule the tight one.

Table F5. Budget sweep with evidence fixed; CrowdHuman unless stated.

K	10	25	50	100	200
Binding share $\Pr(H > K)$ (%)	83	44	15	2.4	.3
Full share $\Pr(\lvert S \rvert = K)$ (%)	75	38	12	2	.2
Admissions (% of $\sum_i Q_i$)	.30	.55	.68	.76	.60
Replacements (% of $\sum_i Q_i$)	1.15	.73	.40	.11	.06
Replacement share (%)	79	57	37	13	9
Refill rate (%)	68	55	38	11	6
Gain over L, CrowdHuman (AP_{50})	1.45	1.28	1.12	.95	.70
Gain over L, COCO-HP (AP)	2.05	1.82	1.62	1.40	1.02

F.7 Seeds

Table F6 lists the five encoder seeds of the focal endpoints. Every seed gains at both budgets, and the seed spread is small against the image-level uncertainty that dominates the intervals of Table 1.

Table F6. Focal CrowdHuman R50-FPN by encoder seed (AP_{50}); L is tuned per seed on development.

Seed	$L, K = 100$	$S, K = 100$	Δ_{100}	Δ_{10}
1	88.19	89.12	.93	1.41
2	88.31	89.27	.96	1.49
3	88.22	89.19	.97	1.44
4	88.29	89.25	.96	1.47
5	88.24	89.17	.93	1.44
Mean	88.25	89.20	.95	1.45
SD	.05	.06	.02	.03

Appendix G Pre-registered analysis specification

The following specification was fixed, dated and committed to the repository before any audit-split measurement; the paper and these appendices report the resulting measurements.

(1) *Primary endpoints:* CrowdHuman focal AP_{50} of S over tuned L at $K = 100$ and at $K = 10$.
(2) *Comparators:* tuned L, greedy maximization of F, and the eleven decoders of Table 2 in both tracks, each with 24 development configurations (Table E2).
(3) *Test families* and their sizes as in Table E3, plus the pooled one-to-one test.
(4) *Risk targets* $(.05, .10)$ on the certified risks; HP monitored.
(5) *Quality variant:* evaluator-matched q ($p_{.5}$ on CrowdHuman and CityPersons, the ten-threshold mean on COCO).
(6) *Prediction* of the focal recovered rate from the other fifteen settings.
(7) *Qualitative scenes:* the COCO low-pressure control of Fig. 2, re-rendered on the same two images from DIAL-Det on Faster R-CNN (Table 1, cross-fitted encoder, $K = 100$), and one budget-binding scene from the focal CrowdHuman R50-FPN, chosen as the audit image with the largest H among those in which, at $K = 10$, S matches an object that L misses.
(8) *Seeds* of the val split, of the cross-fitting halves and of the five encoder runs.

Deviations from this specification are recorded here with their date and reason. 22 September 2026: the budget-binding CrowdHuman scene of item (7) is omitted for space; the paper shows only the COCO low-pressure control of Fig. 2.

Appendix H Provenance and reproducibility

Value ledger. Every experimental value of the paper and of these appendices is a measured output consolidated by `tools/dial_values.py`. The ledger stops when an identity or an ordering fails: the nonnegative terms of (1) and $|G| \geq H \geq Q \geq C \geq C_{\text{ev}}$ for the native decoder, L and S; the factorial's full cell against Table 1; the mass-loss removal against the mass-loss contrast; the oracle's first bar and the corruption curves' origins against the focal shortfall; the status shares summing to one; the stage times summing to the total; the $K = 2$ boundary of Proposition 2(b) against full enumeration and against the closed form (C.3). It warns when a reportable outcome changes — a risk bound above its target, a leave-focal-out prediction outside its 95% prediction interval, or any qualitative statement of Sec. 3 of the paper or of Appendix F becoming false — and regenerates every table and plot of both documents.

Release. The repository contains the candidate exporter, the encoder, the allocator and the independent checker; the configuration of every setting and every decoder, with the grids of Table E2; the per-image records of Table E1 for all settings and decoders; the certificate records of the focal audit (Table D1); the dated analysis specification of Appendix G; and the scripts that regenerate every table and figure of the paper and of these appendices from the records.

Exact and quoted values. Analytic panels (Fig. 1c,d; Fig. C1), closed forms and worked examples are exact. Published results quoted for context in the paper (90.4–93.8 AP_{50} and 39.7–41.9 MR for crowd-retrained detectors) are taken from the cited papers and are not matched reruns.

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